Homework:

1) $a_n \frac{1}{n(n+1)(n+3)(n+4)} = \frac{n+2}{n(n+1)(n+2)(n+3)(n+4)}$

$$\frac{n+4-2}{n(n+1)(n+2)(n+3)(n+4)} = \frac{1}{n\underbrace{(n+1)(n+2)(n+3)}_{b_n}} - \frac{2}{n\underbrace{(n+1)(n+2)(n+3)(n+4)}_{c_n}}$$

$$b_n = \frac{1}{n(n+1)(n+2)(n+3)} = \frac{1}{3}\left(\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}\right)$$

$$\Rightarrow a_{1n} = \frac{1}{3}\sum_{k=1}^{n}\left(\frac{1}{k(k+1)(k+2)} - \frac{1}{(k+1)(k+2)(k+3)}\right) = \frac{1}{3}\left(\frac{1}{1\cdot2\cdot3} - \frac{1}{2\cdot3\cdot4}\right.$$

$$\left. + \frac{1}{2\cdot3\cdot4} - \frac{1}{3\cdot4\cdot5} + \ldots + \frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}\right)$$

$$= \frac{1}{3}\left(\frac{1}{6} - \frac{1}{(n+1)(n+2)(n+3)}\right)$$

$$a_1 = \lim a_{1n} = \frac{1}{18}$$

$$c_n = \frac{2}{4}\left(\frac{n+4-n}{n(n+1)(n+2)(n+3)(n+4)}\right) = \frac{1}{2}\left(\frac{1}{n(n+1)(n+2)(n+3)} - \right.$$

$$\left. \frac{1}{(n+1)(n+2)(n+3)(n+4)}\right) \Rightarrow$$

$$a_{2n} = \frac{1}{2}\left(\frac{1}{1\cdot2\cdot3\cdot4} - \frac{1}{2\cdot3\cdot4\cdot5} + \ldots + \frac{1}{n(n+1)(n+2)(n+3)} - \frac{1}{(n+1)\ldots(n+4)}\right)$$

$$\Rightarrow a_2 = \lim a_{2n} = \frac{1}{2}\cdot\frac{1}{24} = \frac{1}{48}$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n = \frac{1}{18} - \frac{1}{48} = \frac{8-3}{144} = \frac{5}{144}$$

2) $a_n = \{ (1+\sqrt{2})^{2n-1} \}$

Let $x_n = (1+\sqrt{2})^{2n-1} - (\sqrt{2}-1)^{2n-1}$

$$x_n = \binom{2n-1}{0} \sqrt{2}^0 1^{2n-1} + \binom{2n-1}{1} \sqrt{2}^1 1^{2n-2} + \dots +$$

$$\binom{2n-1}{2n-1} \sqrt{2}^{2n-1} 1^0 - \left(-\binom{2n-1}{0} \sqrt{2}^0 1^{2n-1} + \binom{2n-1}{1} \sqrt{2}^1 1^{2n} - \dots - \binom{2n-1}{2n-2} \sqrt{2}^{2n-2} 1^1 + \binom{2n-1}{2n-1} \sqrt{2}^{2n-1} 1^0 \right)$$

$$x_n = 2\left(\binom{2n-1}{0} \sqrt{2}^0 1^{2n-1} + \dots + \binom{2n-1}{2n-2} \sqrt{2}^{2n-2} 1^1 \right)$$

$x_n \in \mathbb{Z} \Rightarrow (\sqrt{2}+1)^{2n-1} = x_n + (\sqrt{2}-1)^{2n-1}$

$$\Rightarrow a_n = \{ x_n + (\sqrt{2}-1)^{2n-1} \} = \{ (\sqrt{2}-1)^{2n-1} \}$$

$\in (0,1) \Rightarrow (\sqrt{2}-1)^{2n-1}$

$$\sum_{n=1}^{\infty} (\sqrt{2}-1)^{2n-1} \overset{G.S}{=} \frac{\sqrt{2}-1}{2\sqrt{2}-2} = \frac{1}{2}$$

$$g = (\sqrt{2}-1)^2 = 3-2\sqrt{2}$$